

A Report on:

Improper Disposal of Electronic Waste

Proposed Solution:

E-Waste Collection & Recycling Portal



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Abstract

Electronic waste, commonly known as e-waste, is one of the fastest-growing waste streams in the world. It includes discarded computers, mobile phones, televisions, printers, refrigerators, batteries, network equipment, electrical appliances and other products containing electronic components. The rapid development of technology, short product replacement cycles, limited repairability and irresponsible disposal have created serious environmental, economic and public-health problems. In 2022, the world generated approximately 62 million tonnes of e-waste, but only 22.3% was documented as formally collected and recycled in an environmentally sound manner. Global e-waste generation is projected to reach approximately 82 million tonnes by 2030.

This report proposes an E-Waste Digital Passport, a digital record assigned to every electronic device during manufacture or sale. The passport would contain information about the device's ownership, specifications, materials, repair history, usage period, refurbishment status and final recycling or disposal process. A QR code, NFC tag or serial number would connect the physical device to its digital record. Consumers, manufacturers, repair centres, recyclers, government authorities and retailers would access different levels of information according to their roles.

The proposed system would improve traceability throughout the device life cycle and reduce the possibility of electronic products being abandoned, illegally dumped or sent to unregistered recyclers. It would encourage repair and reuse, support extended producer responsibility, improve recycling data and help recover valuable materials. The solution is aligned with Sustainable Development Goal 12, particularly responsible consumption, production and waste reduction. It also supports the life-cycle perspective promoted by ISO 14001 and the life-cycle assessment framework described by ISO 14040.

The report analyses the problem of improper e-waste disposal, explains the architecture and operation of the proposed passport, evaluates its sustainability, feasibility and environmental benefits, and identifies the challenges associated with privacy, cost, data accuracy, cybersecurity and adoption.

1. Executive Summary

Electronic waste (e-waste) is a rapidly growing environmental challenge caused by increasing electronics consumption, short replacement cycles and limited visibility into what happens to devices after use. The Global E-waste Monitor 2024 reported 62 million tonnes of e-waste generated globally in 2022 and projected 82 million tonnes by 2030 if current trends continue. [1]

This report proposes an E-Waste Digital Passport, a web-based system that gives an electronic device a persistent digital identity and records important lifecycle information. Instead of treating disposal as a single event, the proposed system follows the device through registration, use, repair, reuse, transfer, collection, refurbishment and recycling. The concept is inspired by the broader Digital Product Passport (DPP) approach. The European Commission describes a DPP as a digital container for product information that can support sustainability, circularity, repair, reuse and recycling while improving traceability. [2] The proposed student prototype adapts these principles specifically to e-waste.

2. Introduction

Electronic products contain valuable materials as well as substances requiring careful handling. When unwanted devices are mixed with ordinary waste, opportunities for reuse and material recovery can be lost. India's Central Pollution Control Board states that e-waste management is regulated by the E-Waste (Management) Rules, 2022, effective from 1 April 2023. [3][4]

Core idea

The system does not simply ask, “Where can I recycle this?” It asks, “What is this device, what has happened to it, what should happen next, and can its final destination be verified?”

2.1 Objectives

- Create a unique digital identity for each registered electronic device.
- Record device condition, ownership/lifecycle events and end-of-life status.
- Encourage repair, reuse and refurbishment before recycling where appropriate.
- Provide QR-based access to a controlled, privacy-aware digital passport.
- Allow verified recyclers to confirm collection and final processing.
- Generate a digital recycling certificate and auditable lifecycle history.

3. Problem Analysis

A conventional e-waste workflow often ends when a consumer gives away, sells, discards or hands over a device. The consumer may not know whether the item was repaired, reused,

dismantled or recycled. This creates a traceability gap. A collection directory can tell a user where to go, and an incentive system can motivate participation, but neither necessarily creates a persistent record of the device itself.

3.1 Key Problems

Problem	Impact	Passport Response
No persistent device identity	Lifecycle events are difficult to connect	Unique passport ID + QR
Poor lifecycle visibility	Repair/reuse opportunities may be missed	Lifecycle timeline
Unclear end-of-life outcome	User cannot verify final destination	Recycler verification + certificate
Scattered product information	Repairers/recyclers lack context	Structured device record
Privacy concerns	Public sharing can expose personal data	Role-based access + public/private views

3.2 Existing Approaches and Differentiation

Existing digital approaches generally focus on one part of the e-waste problem: locating collection points, arranging pickups, offering incentives, or listing devices for resale. The proposed E-Waste Digital Passport is differentiated by making the electronic device—not the collection point or reward—the primary digital object.

Approach	Primary Function	Limitation	Passport Difference
Collection locator	Find facilities	Does not track device history	Tracks individual device
Reward platform	Incentivize recycling	Reward is central	Lifecycle record is central
Marketplace	Sell/donate	Often ends at transaction	Continues to end-of-life
Collection scheduler	Book pickups	Focuses on logistics	Adds post-collection traceability

3.3 Design Principle

The system follows a circular hierarchy: extend useful life first, then recover materials when reuse is no longer practical. This supports the broader circular-economy principle that products and materials should remain useful for as long as possible.

4. Proposed System: E-Waste Digital Passport

The E-Waste Digital Passport is a role-based web application. A user registers an electronic device and receives a unique passport identifier. The passport stores structured information about the device and its lifecycle. Authorized users can append events, while sensitive information remains restricted.

4.1 Functional Requirements

1. User registration and secure login.
2. Device registration with category, brand, model, age, condition and optional serial/IMEI information.
3. Automatic generation of a unique passport identifier.
4. Generation of a QR code/data carrier linking to the passport.
5. Lifecycle-event creation for repair, transfer, donation, refurbishment, collection and recycling.
6. User dashboard showing registered devices and current status.
7. Recycler/admin dashboard for verification and status updates.
8. Public passport view containing only non-sensitive information.
9. Digital recycling certificate after verified recycling.
10. Audit log of important changes.

4.2 Non-Functional Requirements

- Usability for non-technical users.
- Authentication, authorization, password hashing and input validation.
- Privacy protection for addresses, phone numbers and other personal data.
- Reliable storage of lifecycle history without silently overwriting past events.
- Scalability for many users/devices.
- Accessible typography, contrast and keyboard-friendly controls.
- Modular frontend and REST-style backend for maintainability.

4.3 User Roles

Role	Main Permissions
Consumer/User	Register devices, view passport, add permitted lifecycle events, view certificates
Recycler	View assigned devices, verify collection/recycling, add processing details
Administrator	Manage users/recyclers, audit records, verify

	organizations
Public Viewer	Scan QR and view limited non-sensitive passport information

5. System Architecture

The proposed prototype uses a three-tier web architecture. The presentation layer provides user and recycler interfaces. The application layer implements business rules and access control. The data layer stores users, devices and lifecycle events.

Layer	Technology / Components	Purpose
Presentation	React + Vite + Tailwind CSS	Dashboard, passport, QR view, recycler UI
Application	Node.js + Express	Authentication, passport service, lifecycle service, verification
Data Access	REST API + parameterized SQL/ORM	Validation, authorization and database operations
Database	MySQL	Users, devices, events, recyclers, certificates
Support	QR library; optional maps/email	Data carrier, notifications and future integrations

5.1 Main Data Flow

- 11. User authenticates and opens Add Device.
- 12. Application validates input and creates the device record.
- 13. A unique passport ID is generated and linked to the device.
- 14. A QR code is generated containing a safe public URL/token.
- 15. User records lifecycle events such as repair or transfer.
- 16. A verified recycler confirms collection or recycling.
- 17. The system creates the final recycling record and certificate.
- 18. An audit log preserves who performed sensitive actions and when.

5.2 Suggested Technology Stack

Component	Technology	Purpose
Frontend	React, Vite, Tailwind CSS	Responsive interface
Backend	Node.js, Express	REST API and business logic
Database	MySQL	Persistent structured data

Authentication	JWT/session + bcrypt/Argon2	Identity and access control
QR	QR-code generation library	Device data carrier
Charts	Recharts	Dashboard analytics

6. Database Design

A relational database is appropriate because the system contains strongly related entities and requires reliable history. Lifecycle events should be stored as separate records rather than repeatedly overwriting the current device state.

6.1 Core Tables

Table	Important Fields	Purpose
users	id, name, email, password_hash, role	Authentication and identity
devices	id, passport_id, user_id, category, brand, model, purchase_year, condition, status	Core device record
device_events	id, device_id, event_type, event_date, actor_id, notes	Lifecycle history
recyclers	id, organization, location, verification_status	Recycler identities
recycling_records	id, device_id, recycler_id, date, weight, method	Verified end-of-life processing
certificates	id, device_id, certificate_no, issued_at	Digital recycling certificate
audit_logs	id, actor_id, action, entity_id, timestamp	Security/accountability

6.2 Example Passport Record

Field	Example
Passport ID	EW-2026-000124
Device	Lenovo IdeaPad 5
Category	Laptop / ICT Equipment
Purchase Year	2022
Condition	Fair
Current Status	Repair Recommended

Public QR URL	/passport/EW-2026-000124
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6.3 Security and Privacy

- Never store passwords in plain text; use strong password hashing.
- Use role-based access control for recycler/admin functions.
- QR codes should expose a public token/passport ID, not a private database identifier.
- Public passport pages should hide addresses, phone numbers and other personal data.
- Lifecycle events should record actor and timestamp.
- Validate input on client and server sides.
- Log sensitive changes such as recycler verification.

7. System Workflow and User Interface

7.1 Consumer Workflow

Stage	System Action
1. Register / Login	User creates account and authenticates
2. Add Device	User enters device and condition details
3. Generate Passport	System creates unique passport ID
4. Generate QR	System links QR/data carrier to public passport
5. Record Lifecycle	User/authorized actor adds repair, reuse or transfer events
6. End of Life	Device enters collection/processing
7. Verify Recycling	Recycler records final processing
8. Issue Certificate	System marks Recycled and generates certificate

7.2 Digital Passport Interface

The passport page should present device identity first, followed by current status and a visual lifecycle timeline. A QR code should link to a privacy-filtered public view. Editing lifecycle events requires authentication and appropriate permissions.

UI Section	Content
Identity	Image, brand/model, passport ID, category
Current status	In Use / Repair / Reuse / Collection / Recycled
Condition	Working state, condition score, last assessment

Timeline	Chronological lifecycle events
QR / Share	Safe public passport link
Certificate	Available after verified recycling

7.3 Recycler Workflow

19. Recycler logs into verified dashboard.
20. Recycler scans or enters passport ID.
21. System displays information permitted for recycler role.
22. Recycler records collection/processing information.
23. System validates fields and records actor + timestamp.
24. Final processing marks device Recycled and issues certificate.

7.4 Example Lifecycle

Example device

Lenovo laptop purchased in 2022 → registered in 2026 → battery repaired in 2027 → transferred to a second user in 2028 → declared end-of-life in 2030 → collected by verified recycler → recycling verified → digital certificate issued.

8. Implementation Approach

8.1 Passport Identifier

A passport identifier should be unique, non-sensitive and easy to communicate. A prototype can use EW-YYYY-NNNNN. The database should enforce uniqueness so two devices cannot receive the same passport ID.

8.2 Lifecycle State Model

State	Meaning	Typical Actor
REGISTERED	Device added to platform	User
IN_USE	Device remains active	User
REPAIR	Device undergoing repair	User / Repairer
REUSED	Device transferred for further use	User
COLLECTED	Device enters formal collection	Recycler
PROCESSING	Device dismantled/refurbished/recycled	Recycler

RECYCLED	Final recycling verified	Recycler/Admin
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8.3 Condition / Circularity Score

A prototype may calculate a simple decision-support score from age, working status, repairability and physical condition. It should be clearly labeled as a prototype metric rather than an official environmental standard.

Example rule

If a device is functional and repairable, recommend REPAIR/REUSE. If it is non-functional and impractical to repair, recommend RECYCLE. The recommendation is informational and does not replace professional assessment.

8.4 QR Code Design

The QR code should point to a stable public route such as `/passport/EW-2026-000124` or, preferably, a random public token. Public pages should expose only approved information; editing requires authentication.

8.5 REST API Endpoints

Method	Endpoint	Purpose
POST	<code>/api/auth/register</code>	Create account
POST	<code>/api/devices</code>	Register device
GET	<code>/api/devices/:id</code>	View authorized device
POST	<code>/api/devices/:id/events</code>	Add lifecycle event
GET	<code>/api/passport/:passportId</code>	Public passport
POST	<code>/api/recycling/:deviceId</code>	Verify recycling
GET	<code>/api/certificates/:deviceId</code>	Retrieve certificate

9. Benefits, SDG Alignment and Feasibility

9.1 Environmental Benefits

- Encourages longer device lifetimes by making repair and reuse visible.
- Improves traceability of devices entering formal recycling.
- Provides structured information that can support sorting and responsible processing.
- Shows recycling as the final stage of a broader product lifecycle.

9.2 Social and Economic Benefits

- Gives consumers a simple record of what happened to their device.
- Can improve trust in formal recycling through verifiable final status.
- Provides repairers and recyclers with structured information.
- Supports circular-economy practices by keeping products in use longer and recovering materials.

9.3 SDG Alignment

SDG	Connection
SDG 12 – Responsible Consumption and Production	Directly supports responsible use, repair, reuse and recycling.
SDG 11 – Sustainable Cities and Communities	Can support organized urban e-waste workflows.
SDG 13 – Climate Action	May reduce impacts associated with unnecessary replacement and material extraction.

The strongest alignment is with SDG 12 because the project directly targets responsible consumption, product lifetime extension and end-of-life management.

9.4 Feasibility

A minimum viable prototype can be implemented without advanced hardware. React can provide the interface, Node.js/Express the APIs, MySQL the persistence layer, and a QR library the data carrier. The first version can use manually entered lifecycle events and later integrate with verified recyclers.

9.5 Expected Prototype Output

- Working login and registration.
- Device registration and generated passport.
- QR code and privacy-filtered public passport.
- Lifecycle timeline and event history.
- Recycler verification dashboard.
- Digital recycling certificate.
- Database schema and REST API.

10. Limitations and Future Scope

10.1 Current Limitations

- A student prototype cannot independently verify every recycler; real deployment should connect to trusted organizations or regulatory data.
- Condition and circularity scores are estimates unless supported by professional inspection.
- Users may enter inaccurate information.
- A QR code alone does not prevent fraud; authentication and audit logs are required.
- Real deployment needs data governance, privacy policies and integration agreements.

10.2 Future Enhancements

- Integration with authorized recycler registries and producer/manufacturer systems.
- AI-assisted device identification and condition assessment.
- Automatic model extraction from photographs or barcodes.
- Mobile application with offline QR scanning.
- Tamper-evident or distributed-ledger records for selected events.
- Integration with repair/refurbishment providers and take-back programs.
- Analytics for institutions and municipalities.
- Interoperability with broader Digital Product Passport ecosystems.

11. Conclusion

Improper disposal of electronic waste is not only a collection problem; it is also a traceability and lifecycle-management problem. The proposed E-Waste Digital Passport addresses this gap by giving each electronic device a persistent digital identity and maintaining a structured history of its use, repair, reuse, transfer, collection and recycling.

The system is technically feasible as a web application and provides a clear path from a college prototype to a larger circular-economy platform. By encouraging products to remain useful for longer and creating a verifiable record when recycling becomes necessary, the system directly supports responsible consumption and production.

References

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